

VIKAS TIWARI

Software Engineer — Backend, Distributed Systems & Applied AI

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EDUCATION

Indian Institute of Information Technology (IIIT), Bhopal

Aug 2023 – Jun 2027

Bachelor of Technology, Information Technology — CGPA: 8.43/10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Full-Stack Web Development, Software Engineering Fundamentals

EXPERIENCE

Competitive Programming Member — CODAME, IIIT Bhopal

Nov 2024 – Sep 2025

- Led 10 workshops on data structures, algorithms, and time complexity analysis for 120 first-year students, establishing CODAME's core onboarding curriculum for new competitive programmers.
- Authored and reviewed 15 contest problems; contributed to a 65% increase in club outreach through cross-team promotional coordination.

PROJECTS

Multi-Modal Product Intelligence Engine [GitHub](#)

Python 3.12, FastAPI, Qdrant, Redis, CLIP, BGE, Sentence Transformers, Docker Compose, Prometheus, pytest

- Engineered a multi-modal catalog platform indexing 512-dimensional CLIP image vectors and 384-dimensional BGE text vectors in Qdrant, serving hybrid search through weighted image-text similarity fusion with optional cross-encoder reranking.
- Designed an asynchronous ingestion pipeline offloading embedding generation to a background worker pool, returning upload responses immediately while Redis holds job queue, product state, recommendation cache, and analytics buckets as the primary datastore (AOF persistence enabled).
- Implemented duplicate detection via weighted-similarity scoring across 3 decision tiers (OFF/WARN/BLOCK) with an explainable cross-encoder verifier, plus deterministic pricing estimation from comparables using trimmed mean, weighted average, and median strategies.
- Added structured decision traces for explainability, Prometheus metrics with health and readiness probes, and an opt-in enterprise layer covering API keys, RBAC, tenant isolation, audit logging, and usage quotas.
- Sustained 1,327 passing tests at 99% branch coverage across 69 commits, gated by CI and pre-commit hooks running ruff, black, and mypy on every push.

RAG Evaluation System [GitHub](#)

Python, FAISS, BM25, FastAPI, AWS Lambda, Sentence Transformers, Groq LLM API

- Built a retrieval-augmented generation pipeline without LangChain, fusing dense FAISS and sparse BM25 retrieval through Reciprocal Rank Fusion (k=60) with cross-encoder reranking over a 384-dimensional BGE embedding space.
- Quantified pipeline quality against a 15-question ground-truth set, reaching 1.00 Recall@5, 1.00 MRR, and 1.00 Hit Rate, measured through a custom harness computing Precision@K, Recall, MRR, and LLM-judged faithfulness.
- Deployed a streaming SSE endpoint on AWS Lambda via Mangum and SAM, benchmarking 0.2 ms BM25 retrieval and 1.5-3.5 s end-to-end latency; validated by 289 tests at 87% coverage across Python 3.11 and 3.12.

Distributed Key-Value Store [GitHub](#)

C++20, TCP Sockets, epoll, POSIX Threads, Write-Ahead Logging, CMake, GoogleTest

- Wrote a networked key-value store from raw TCP sockets, pairing a select-based server with an edge-triggered epoll event loop and a shared_mutex thread pool that sustained 1,000 concurrent connections.
- Measured 9,619 ops/sec on a single connection and 4,831 ops/sec at 1,000 concurrent clients — a 16% degradation — with p99 latency of 4.62 ms, using a reproducible harness sweeping 7 concurrency levels.
- Guaranteed durability through append-only write-ahead logging with periodic snapshots and atomic file rename, recovering 10,000 keys in 410 ms after SIGKILL with zero data loss.

ACHIEVEMENTS

- Solved 500+ DSA problems across LeetCode (91st percentile, peak rating 1783), Codeforces (Specialist, peak rating 1410), and CodeChef (4-Star, peak rating 1828).
- Ranked 161st globally out of 35,000+ participants in a CodeChef Starters contest.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

Backend & APIs: FastAPI, Node.js, Express.js, REST API Design, Asynchronous Processing, Background Workers, JWT/OAuth, RBAC

AI & Retrieval: RAG, Vector Databases (Qdrant, FAISS), Multi-Modal Embeddings (CLIP, BGE), Hybrid Search, Reciprocal Rank Fusion, Cross-Encoder Reranking, BM25, Sentence Transformers, LLM APIs, Prompt & Context Engineering

Data & Infrastructure: Redis, PostgreSQL, MongoDB, MySQL, Docker, Docker Compose, Git, GitHub Actions (CI/CD), AWS Lambda, Prometheus, Linux

Frontend: React.js, Redux, Tailwind CSS, Vite | **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Engineering Practices: Test-Driven Development, pytest, GoogleTest, Static Typing (mypy), Linting (ruff, black), Benchmarking, Observability, Concurrency, TCP/IP Socket Programming